

Pointwise extensions and sketches in bicategories

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August 15, 2013

2010 Mathematics Subject Classification: 18D10; 18D05

Key words and phrases: bicategory; pointwise Kan extension; right lifting; cofibration; collage; lax limit.

Abstract

We make a few remarks concerning pointwise extensions in a bicategory which include the case of bicategories of enriched categories. We show that extensions, pointwise or not, can be replaced by extensions along very special fully faithful maps. This leads us to suggest a concept of limit sketch internal to the bicategory.

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*The author gratefully acknowledges the support of Australian Research Council Discovery Grant DP130101969.

1 Extensions and liftings in a bicategory

We begin by recalling the concepts of extension and lifting as used, for example, in [10] and [15]. Let $\mathcal{M}$ be a bicategory. Consider a triangle

$$\begin{array}{ccc} A & \xrightarrow{j} & B \\ & \searrow f & \swarrow r \\ & X & \end{array} \quad \begin{array}{c} \xleftarrow{\rho} \end{array} \quad (1.1)$$

in $\mathcal{M}$. We say the 2-morphism ρ in (1.1) exhibits r as a *right extension* of f along j when pasting the triangle to 2-morphisms $\theta : g \Rightarrow r$ provides a bijection between such 2-morphisms θ and 2-morphisms $\phi : gj \Rightarrow f$. Then r is unique up to a unique isomorphism. We write $\text{hom}_A(j, f)$ for a right extension of $f : A \rightarrow X$ along $j : A \rightarrow B$. To say every f has a right extension along j is to say the functor

$$\mathcal{M}(j, 1) : \mathcal{M}(B, X) \rightarrow \mathcal{M}(A, X) ,$$

defined by precomposition with j , has a right adjoint ran_j ; so $\text{ran}_j f = \text{hom}_A(j, f)$. If also $i : B \rightarrow C$ then

$$\text{hom}_A(ij, f) \cong \text{hom}_B(i, \text{hom}_A(j, f)) . \quad (1.2)$$

Left extensions in $\mathcal{M}$ are right extensions in $\mathcal{M}^{\text{co}}$: we write $\text{lan}_j f$ for a left extension of f along j .

Right liftings in $\mathcal{M}$ are right extensions in $\mathcal{M}^{\text{op}}$. We write $\text{hom}^B(m, n)$ for a right lifting of $n : X \rightarrow B$ through $m : A \rightarrow B$. If also $\ell : C \rightarrow A$ then

$$\text{hom}^B(m\ell, n) \cong \text{hom}^A(\ell, \text{hom}^B(m, n)) . \quad (1.3)$$

A *left preadjoint* for $n : X \rightarrow B$ is a right extension $\text{hom}_X(n, 1_X)$ of the identity 1-morphism 1_X of X along n . Left adjoints are particular left preadjoints: when $j \dashv j^*$, we have

$$j \cong \text{hom}_A(j^*, 1_A) .$$

Left adjoints are sometimes called *maps*. A map $j : A \rightarrow B$ with right adjoint $j^* : B \rightarrow A$ is *fully faithful* when the unit $\eta_j : 1_A \Rightarrow j^*j$ is invertible.

2 Pointwise extensions

Suppose $\mathcal{K}$ is a class of maps of $\mathcal{M}$ which is closed under composition and includes the identities. We regard $\mathcal{K}$ as a locally full sub-bicategory of $\mathcal{M}$. When regarding a morphism f of $\mathcal{K}$ as in $\mathcal{M}$, we write it as f_* . This situation was studied by Richard Wood [16, 17].

Definition. If (1.1) is in $\mathcal{K}$, we say it is a *pointwise right extension* in $\mathcal{K}$ when it is a right extension in $\mathcal{M}$. So

$$r \cong \text{hom}_A(j_*, f) . \quad (2.4)$$

With this definition, the first proposition is obvious.

Proposition 1. *If (1.1) is a pointwise right extension in $\mathcal{K}$ then it is a right extension in $\mathcal{M}$.*

Proposition 2. *If (1.1) is a pointwise right extension in $\mathcal{K}$ and j is fully faithful then ρ is invertible.*

Proof. The right extension of r through j^* is rj . Using (1.2), we see that rj is the right extension of f through $j^*j_* \cong 1_A$. The right extension through an identity is the same 1-morphism. So $rj \cong f$. One of the adjunction identities for $j_* \dashv j^*$ shows that this isomorphism is ρ . $\square$

A 1-morphism $j : B \rightarrow C$ is called a *cofibration* when each $\mathcal{M}(j, X) : \mathcal{M}(C, X) \rightarrow \mathcal{M}(B, X)$ is a fibration in the bicategorical sense of [13]. A 1-morphism $i : A \rightarrow C$ is called a *coopfibration* when each $\mathcal{M}(i, X) : \mathcal{M}(C, X) \rightarrow \mathcal{M}(A, X)$ is an opfibration in the bicategorical sense of [13].

If $u : A \rightarrow B$ is a 1-morphism of $\mathcal{K}$, an object X is *u-complete* when every $f : A \rightarrow X$ has a pointwise right extension along u . In that case, if $\mathcal{M}$ admits right extensions, the adjunction

$$\mathcal{M}(u_*, 1) \dashv \text{hom}_A(u_*, -) : \mathcal{M}(A, X) \rightarrow \mathcal{M}(B, X)$$

restricts to an adjunction

$$\mathcal{K}(u, 1) \dashv \text{ran}_u : \mathcal{K}(A, X) \rightarrow \mathcal{K}(B, X) .$$

A *collage* or *lax colimit* [14] of $j : A \rightarrow B$ in $\mathcal{M}$ is a universal 2-morphism

$$\begin{array}{ccc} A & \xrightarrow{j} & B \\ & \swarrow i_1 & \searrow i_0 \\ & C(j) & \end{array} \quad \text{with } \nu : i_0 \circ j \Rightarrow i_1 \quad (2.5)$$

That is, pasting with ν determines an equivalence of categories

$$\mathcal{M}(C(j), X) \simeq \mathcal{M}(j, X) / \mathcal{M}(A, X) , \quad (2.6)$$

where the slice category on the right-hand side is the comma category [8] of the functor $\mathcal{M}(j, X)$ and the identity functor of $\mathcal{M}(A, X)$.

Proposition 3. *In the collage (2.5), the 1-morphism i_0 is a map with invertible unit $1_B \cong i_0^* i_0$, and with $i_0^* i_1 \cong j$ compatibly with ν . If $\mathcal{M}$ has local initial objects, preserved by composition with each given 1-morphism, and if i_1 is a map, then the unit $1_A \cong i_1^* i_1$ is invertible. Furthermore, i_0 is a cofibration and i_1 is a coopfibration.*

Examples in [14] suggest consideration of the following properties of a collage.

Definition. We say that a collage (2.5) is $\mathcal{K}$ -compatible when

1. i_0 and i_1 are in $\mathcal{K}$,
2. for any $h : C(j) \rightarrow X$, if hi_0 and hi_1 are in $\mathcal{K}$, so is h , and
3. the mate triangle

$$\begin{array}{ccc}
 & C(j) & \\
 i_1^* \swarrow & \hat{\nu} & \searrow i_0^* \\
 A & \xrightarrow{j} & B
 \end{array} \tag{2.7}$$

exhibits $C(j)$ as a lax limit of j in $\mathcal{M}$.

From the universal property (2.6), the triangle (1.1) induces a 1-morphism $\langle r, \rho \rangle : C(j) \rightarrow X$ and invertible 2-cells $f \cong \langle r, \rho \rangle i_1$ and $r \cong \langle r, \rho \rangle i_0$ which conjugate $\langle r, \rho \rangle \nu$ to give ρ .

Now consider the triangle

$$\begin{array}{ccc}
 A & \xrightarrow{i_1} & C(j) \\
 f \searrow & \xleftarrow{\cong} & \swarrow \langle r, \rho \rangle \\
 & X &
 \end{array} \tag{2.8}$$

Proposition 4. *In $\mathcal{M}$, triangle (2.8) exhibits $\langle r, \rho \rangle$ as a right extension of f along i_1 if and only if triangle (1.1) exhibits r as a right extension of f along j .*

Proof. There is a right extension

$$\begin{array}{ccc}
 C(j) & \xrightarrow{i_0^*} & B \\
 \langle r, \rho \rangle \searrow & \xleftarrow{\langle r, \rho \rangle \varepsilon} & \swarrow \langle r, \rho \rangle i_0 \\
 & X &
 \end{array} \tag{2.9}$$

coming from the adjunction $i_0 \dashv i_0^*$. If (2.8) is a right extension, we just paste the right extension (2.9) to the right side of it and use the isomorphisms $i_0^* i_1 \cong j$ and $\langle r, \rho \rangle i_0 \cong r$ to see that (1.1) is a right extension.

Conversely, take $h : C(j) \rightarrow X$ and $\theta : hi_1 \Rightarrow f$. By the universal property (2.6) of $C(j)$, in order to give a 2-morphism $h \Rightarrow \langle r, \rho \rangle$, we precisely require two

2-morphisms $\theta : hi_1 \Longrightarrow f$ and $\phi : hi_0 \Longrightarrow r$ such that $\theta(h\nu) = \rho(\phi j)$. However, each θ determines such a unique ϕ by the right extension property of (1.1). So (2.8) is a right extension. $\square$

Now by applying Proposition 4 to triangle (1.1) thought of as in $\mathcal{M}^{\text{coop}}$, we obtain a pointwise version.

Corollary 5. *Suppose the triangle (1.1) is in $\mathcal{K}$. If (2.8) exhibits $\langle r, \rho \rangle$ as a pointwise right extension of f along i_1 then (1.1) exhibits r as a pointwise right extension of f along j . If the collage (2.5) is $\mathcal{K}$ -compatible then the converse also holds.*

Definition. A square

$$\begin{array}{ccc} S & \xrightarrow{p} & U \\ q \downarrow & \xleftarrow{\lambda} & \downarrow b \\ A & \xrightarrow{j} & B \end{array} \quad (2.10)$$

in $\mathcal{K}$ is called *BC* or, by René Guitart [4], *exact* when its mate square

$$\begin{array}{ccc} S & \xrightarrow{p} & U \\ q^* \uparrow & \xrightarrow{\bar{\lambda}} & \uparrow b^* \\ A & \xrightarrow{j} & B \end{array} \quad (2.11)$$

in $\mathcal{M}$ has $\bar{\lambda}$ invertible.

Proposition 6. *If the triangle (1.1) is a pointwise right extension and the square (2.10) is BC then the 2-morphism obtained by pasting the square on top of the triangle exhibits rb as a right extension of fj along p .*

Proof. Take $h : U \rightarrow X$. Then $\theta : hp \Longrightarrow fq$ are in bijection with $pq^* \Longrightarrow h^*f$ which, by the BC property, are in bijection with $b^*j \Longrightarrow h^*f$, and so with $hb^*j \Longrightarrow f$. By the pointwise right extension property of (1.1), these are in bijection with $hb^* \Longrightarrow r$, and so, by mates, with $\phi : h \Longrightarrow rb$. Then λ , ρ and ϕ paste to give back θ , as required. $\square$

3 The enriched category example

Let $\mathcal{V}$ be a base monoidal category as used in [5]. For two-sided modules (also called bimodules, profunctors and distributors) refer to [7], [14] and [2], for example.

Let $\mathcal{V}\text{-Mod}$ be the bicategory whose objects are $\mathcal{V}$ -categories $\mathcal{A}$ (which are small relative to $\mathcal{V}$), whose 1-morphisms are two-sided modules $M : \mathcal{A} \rightarrow \mathcal{B}$ and whose 2-morphisms are module morphisms. We have an equivalence of categories

$$\mathcal{V}\text{-Mod}(\mathcal{A}, \mathcal{B}) \simeq [\mathcal{B}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]. \quad (3.12)$$

The composite $N \circ M$ of modules $M : \mathcal{A} \longrightarrow \mathcal{B}$ and $N : \mathcal{B} \longrightarrow \mathcal{C}$ is defined by the coend formula:

$$(N \circ M)(C, A) = \int^B m(B, A) \otimes n(C, B) . \quad (3.13)$$

Notice that the hom categories (3.12) are cocomplete and composition is separately colimit preserving (local cocompleteness). Indeed, right liftings and right extensions all exist in $\mathcal{V}\text{-Mod}$. In particular, we have

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ & \swarrow N & \searrow \text{hom}_{\mathcal{A}}(M, N) \\ & \mathcal{X} & \end{array} \quad (3.14)$$

where

$$\begin{aligned} \text{hom}_{\mathcal{A}}(M, N)(X, B) &= \int_A [M(B, A), N(X, A)] \\ &\cong [\mathcal{A}^{\text{op}}, \mathcal{V}](M(B, -), N(X, -)) . \end{aligned} \quad (3.15)$$

Each $\mathcal{V}$ -functor $F : \mathcal{A} \longrightarrow \mathcal{B}$ determines modules $F_* : \mathcal{A} \longrightarrow \mathcal{B}$ and $F^* : \mathcal{B} \longrightarrow \mathcal{A}$ defined by $F_*(B, A) = \mathcal{B}(B, FA)$ and $F^*(A, B) = \mathcal{B}(FA, B)$. In $\mathcal{V}\text{-Mod}$, we have an adjunction $F_* \dashv F^*$; that is, the F_* are some of the maps in $\mathcal{V}\text{-Mod}$.

We identify the 2-category $\mathcal{V}\text{-Cat}$ of $\mathcal{V}$ -categories, $\mathcal{V}$ -functors and $\mathcal{V}$ -natural transformations with the sub-bicategory of $\mathcal{V}\text{-Mod}$ by restricting to modules of the form $F_* \dashv F^*$.

So in Section 1, if we take $\mathcal{M} = \mathcal{V}\text{-Mod}$ then $\mathcal{V}\text{-Cat}$ provides us with a suitable sub-bicategory $\mathcal{K}$.

In this context, a triangle

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{J} & \mathcal{B} \\ & \swarrow F & \searrow R \\ & \mathcal{X} & \end{array} \quad (3.16)$$

in $\mathcal{V}\text{-Cat}$ exhibits R as a pointwise right extension of F along J in the sense of Section 2 if and only if it does in the usual sense for enriched category theory (see [3] or [5]). That is,

$$R_* \cong \text{hom}_{\mathcal{A}}(J_*, F_*) , \quad (3.17)$$

or

$$\mathcal{X}(X, RB) \cong [\mathcal{A}^{\text{op}}, \mathcal{V}](\mathcal{B}(B, J-), \mathcal{X}(X, F-)) . \quad (3.18)$$

The collage

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ & \swarrow i_1 & \searrow i_0 \\ & C(M) & \end{array} \quad (3.19)$$

of a module M is constructed as follows (see [13]). The $\mathcal{V}$ -category $C(M)$ contains $\mathcal{A}$ and $\mathcal{B}$ as disjoint full sub- $\mathcal{V}$ -categories with inclusion $\mathcal{V}$ -functors i_1 and i_0 . There are no other objects. We have

$$C(M)(i_0B, i_1A) = M(B, A)$$

and

$$C(M)(i_1A, i_0B) = 0 ,$$

where 0 is initial in $\mathcal{V}$. Composition is obtained from composition in $\mathcal{A}$ and $\mathcal{B}$, and from their actions on M . This collage is $\mathcal{V}$ -Cat-compatible; see [14] for the lax limit property coming from local cocompleteness of $\mathcal{V}\text{-Mod}$.

Given a span $\mathcal{A} \xleftarrow{Q} \mathcal{S} \xrightarrow{P} \mathcal{U}$ in $\mathcal{V}\text{-Cat}$, it follows easily that its cocomma square is

$$\begin{array}{ccc} \mathcal{S} & \xrightarrow{P} & \mathcal{U} \\ Q \downarrow & \xleftarrow{\lambda} & \downarrow i_0 \\ \mathcal{A} & \xrightarrow{i_1} & C(P_* \circ Q^*) \end{array} \quad (3.20)$$

where λ is the mate of the ν in the collage.

As a simple application of the coend form of the enriched Yoneda Lemma (see [5]), we have the following.

Proposition 7. *Cocomma squares (3.20) are BC.*

Proof. Put $\mathcal{B} = C(P_* \circ Q^*)$. By Yoneda we have

$$i_0^* \circ i_{1*}(U, A) = \int^B \mathcal{B}(B, i_1A) \otimes \mathcal{B}(i_0U, B) \cong \mathcal{B}(i_0U, i_1A) = P_* \circ Q^*(U, A) ,$$

as required. $\square$

For ordinary categories, we have a dual which explains the definition of point-wise right extension used in [10].

Proposition 8. *For $\mathcal{V} = \text{Set}$, comma squares are BC.*

Proof. Consider the comma square

$$\begin{array}{ccc} B/J & \xrightarrow{D_0} & \mathcal{U} \\ D_1 \downarrow & \xleftarrow{\lambda} & \downarrow B \\ \mathcal{A} & \xrightarrow{J} & \mathcal{B} . \end{array} \quad (3.21)$$

The mate of λ is the 2-morphism $D_{0*} \circ D_1^* \longrightarrow B^* \circ J_*$ with component

$$\int^{(V, \beta: BV \rightarrow JX, X)} \mathcal{A}(X, A) \times \mathcal{U}(U, V) \longrightarrow \mathcal{B}(BU, JA)$$

at (U, A) the function whose composite with the injection $\text{in}_{(V, \beta, X)}$ at the object $(V, \beta : BV \rightarrow JX, X)$ of B/J is the function

$$\mathcal{A}(X, A) \times \mathcal{U}(U, V) \longrightarrow \mathcal{B}(BU, JA)$$

taking $(\alpha : X \rightarrow A, v : U \rightarrow V)$ to the composite

$$BU \xrightarrow{Bv} BV \xrightarrow{\beta} JX \xrightarrow{J\alpha} JA .$$

By a familiar argument with tensor products, the function

$$\mathcal{B}(BU, JA) \longrightarrow \int^{(V, \beta : BV \rightarrow JX, X)} \mathcal{A}(X, A) \times \mathcal{U}(U, V) ,$$

taking $\beta : BU \rightarrow JA$ to $\text{in}_{(U, \beta, A)}(1_U, 1_A)$, gives an inverse to the mate of λ . $\square$

4 Sketches

Definition. A *sketch* $\mathbb{T}$ on an object C is a triangle

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ & \swarrow v & \nwarrow w \\ & C & \end{array} \quad \begin{array}{c} \xleftarrow{\tau} \\ \end{array} \quad (4.22)$$

in $\mathcal{K}$ where u is a fully faithful coopfibration and τ is invertible.

A *model* of $\mathbb{T}$ in X is a 1-morphism $f : C \rightarrow X$ in $\mathcal{K}$ such that $f\tau$ exhibits fw as a pointwise right extension of fv along u . In particular, $f\tau$ is invertible. Write $\text{Mdl}(\mathbb{T}, X)$ for the full subcategory of $\mathcal{K}(C, X)$ consisting of the these models.

Proposition 9. *If X is u -complete then $\text{Mdl}(\mathbb{T}, X)$ is the inverter of the natural transformation*

$$\overline{\mathcal{K}(\tau, 1)} : \mathcal{K}(w, 1) \Longrightarrow \text{ran}_u \mathcal{K}(v, 1) : \mathcal{K}(C, X) \longrightarrow \mathcal{K}(B, X)$$

which is the mate of $\mathcal{K}(u, 1)\mathcal{K}(w, 1) \cong \mathcal{K}(wu, 1) \xrightarrow{\mathcal{K}(\tau, 1)} \mathcal{K}(v, 1)$.

Now suppose $\mathcal{M}$ is a monoidal bicategory and that $\mathcal{K}$ is closed under the monoidal structure: that is, if $f : A \rightarrow B$ and $g : C \rightarrow D$ are 1-morphisms of $\mathcal{K}$ then $f \otimes g : A \otimes C \rightarrow B \otimes D$ is in $\mathcal{K}$. Moreover, we suppose $\mathcal{K}$ is closed with internal hom $[A, B]$.

We say that X is *pointwise u -complete* when it is $(1_K \otimes u)$ -complete for every object K .

Notice that, for 1-morphisms u and h in $\mathcal{K}$, we have a BC square:

$$\begin{array}{ccc} H \otimes A & \xrightarrow{1 \otimes u} & H \otimes B \\ h \otimes 1 \downarrow & \xRightarrow{\cong} & \downarrow h \otimes 1 \\ K \otimes A & \xrightarrow{1 \otimes u} & K \otimes B . \end{array} \quad (4.23)$$

It follows that, if X is pointwise u -complete, then

$$\text{ran}_u : \mathcal{K}(K \otimes A, X) \longrightarrow \mathcal{K}(K \otimes B, X)$$

is pseudonatural in K and so transports to a pseudonatural transformation

$$\mathcal{K}(K, [A, X]) \longrightarrow \mathcal{K}(K, [B, X]) .$$

By the bicategorical Yoneda lemma [13], this last transformation is induced by a 1-morphism

$$r_u : [A, X] \longrightarrow [B, X]$$

which is right adjoint to

$$[u, 1] : [B, X] \longrightarrow [A, X]$$

in $\mathcal{K}$.

Proposition 10. *If X is pointwise u -complete then the inverter $\text{mdl}(\mathbb{T}, X)$ of the 2-morphism*

$$\overline{[\tau, 1]} : [w, 1] \Longrightarrow r_u[v, 1] : [C, X] \longrightarrow [B, X] ,$$

which is the mate of $[u, 1][w, 1] \cong [wu, 1] \xRightarrow{[\tau, 1]} [v, 1]$, satisfies the pseudonatural equivalence

$$\text{Mdl}(\mathbb{T}, [K, X]) \simeq \mathcal{K}(K, \text{mdl}(\mathbb{T}, X)) .$$

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